Matt Gottsacker

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Research Interests

Spatial Computing / XR, Human-Computer Interaction, Interruptions, Cross-Reality Interactions and Transitions, 3D User Interfaces, Generative AI.

Skills and Experience

Software prototyping (Unity), Hardware prototyping, Generative AI pipeline design and implementation (text, image), Experiment design, Quantitative and Qualitative analysis.

Education

PhD **University of Central Florida**, Computer Science — Aug 2020 – present

- Expected graduation: Oct 2026. Advisor: Prof. Gregory Welch.
- Dissertation: *XR World Switching: User Interfaces and Cognitive Complexities*.

BS **Saint Louis University**, Computer Science — Aug 2015 – May 2019

- Minor in English (concentration on Rhetoric, Writing, and Technology).

Webster University Vienna, Semester abroad in Vienna, Austria — Jan 2017 – May 2017

Professional Positions and Affiliations

Graduate Research Assistant, Synthetic Reality Lab (SREAL), Institute for Simulation and Training, University of Central Florida — Orlando, FL, Aug 2020 – present

- **PhD Research.** Create novel interfaces for facilitating task transitions and interactions across different Mixed Reality environments. Conduct mixed-methods human-subjects experiments. Analyze results and share findings through peer-reviewed academic publications, conference presentations, and public social media posts. Advisor: Prof. Greg Welch.
- **LLM Pipelines for Research Synthesis.** Designed, implemented, and built multi-agent LLM pipelines (LangChain, Ollama) for large-scale literature synthesis (1000+ papers), along with LLM features and a conversational voice UI in a custom annotation tool to improve speed and quality of research-paper review.
- **Virtual Experience Research Accelerator (VERA).** Develop business models, advise on researcher-centered design/evaluation of platform, lead initial research studies using platform.

Research Intern, Global Technology Applied Research, JPMorganChase — New York, NY, Jun 2023 – Dec 2024

- Formed connections and engaged with company stakeholders to identify needs that could be addressed with emerging technology. Designed and implemented novel prototypes of immersive presentation systems. Conducted user studies to quantitatively and qualitatively evaluate users' experiences. Internship outputs include a CHI 2025 paper, two workshop publications, a pending patent, and an ISMAR 2026 paper currently under review. Mentored and managed by Dr. Mengyu Chen and Dr. Blair MacIntyre.

Visiting Researcher, Computer Graphics and User Interfaces Lab, Columbia University — New York, NY, May 2022 – Aug 2022

- Researched techniques for transitioning among multiple perspectives in a collaborative Mixed Reality virtual environment, and explored interfaces to resolve mismatches between AR users' physical and virtual positions after transitioning into another virtual perspective. Advised by Prof. Steven Feiner.

Alumni Service Corps Teacher, Department of Mathematics, Marquette University High School — Milwaukee, WI — Aug 2019 – Jun 2020

- Taught Computer Programming course using the Java programming language; designed curriculum to be interdisciplinary, creative, innovative, collaborative, and ethical. Mentored students on the FIRST Robotics Competition programming subteam. Served as a Kairos retreat leader. Co-directed the 2019 production of Senior Follies, a play written by students.

Data Science Research Intern, Nintex — Bellevue, WA — Jun 2019 – Aug 2019

- Developed a fully automated data analysis pipeline to gain insights about user behavior. The pipeline detects fresh data and transforms meaningful information into reports for company executives that summarize customer usage and highlight interesting or outlier activity.

Research Intern, MIT Lincoln Laboratory — Lexington, MA — May 2018 – Jul 2018

- Worked on a Software-Defined Networking application that allows network analysts to write access control policies based on high-level identifiers like usernames that are enforced at the network level. Created a web-based application for the system to visualize the bindings between pairs of network identifiers.

Awards and Honors

Dagstuhl Seminar 27151 Invitee: Beyond XR. Selected for the invitation-only, week-long research seminar at Schloss Dagstuhl — Leibniz Center for Informatics (Germany), convening leading international researchers on the convergence of extended reality with body-worn monitoring, perception, and adaptive human-computer interaction. — Apr 2027

Meta PhD Research Fellowship Finalist. One of two finalists in the AR/VR Future Technologies program track. Over 2,300 applicants across 21 tracks. — 2022

Doctoral Consortium at *IEEE Virtual Reality and 3D User Interfaces (VR)*. Selected to present and discuss dissertation plans with AR/VR senior researchers. — 2022

James D. Collins Award for Academic Excellence at *Saint Louis University*. Recognized by the Computer Science department for outstanding scholastic and creative achievement; awarded to one senior each year. — 2019

Intern Idea Innovation Challenge Winner at *MIT Lincoln Laboratory*. With four other interns, designed a system using natural language processing to analyze workplace speech patterns and inform employees of implicit biases. Presented to five MIT LL executives. — 2018

Deloitte Consulting Challenge Winner at *Saint Louis University*. Designed and presented a mixed-use urban redevelopment plan for Midtown in St. Louis. — 2018

PricewaterhouseCoopers Consulting Challenge Winner at *Saint Louis University*. Designed and presented case study convenience store point-of-sale automation solution. — 2018

Knoedler Student Research Grant. Travel scholarship to attend the *ACM Internet Measurement Conference* at Northeastern University in Boston, MA. — 2018

GENI Regional Workshop Travel Grant. Travel scholarship to attend workshop about the Global Environment for Network Innovations at the University of Oregon. — 2017

Presidential Scholarship Finalist at *Saint Louis University*. Scholarship for exemplary student leaders, totaling $80,000 over four years. — 2015

Patents

Matt Gottsacker, Mengyu Chen, David Saffo, Feiyu Lu, Blair MacIntyre. United States Provisional Patent Pub. No. 2025/0022020 A1: **Systems and Methods for Audience Feedback Guided Mixed Reality** (Pending).

Gregory Welch, **Matt Gottsacker**, Nahal Norouzi, Gerd Bruder. United States Provisional Patent Pub. No. 2023/0336804 A1: **Contextual Audiovisual Synthesis for Attention State Management** (Pending).

Gregory Welch, **Matt Gottsacker**, Nahal Norouzi, Gerd Bruder. United States Patent No. 11,729,448: **Intelligent Digital Interruption Management**.

Publications

★ *indicates a publication I am particularly proud of or excited about.*

Journal Publications

[1] Robbe Cools, Inne Maerevoet, **Matt Gottscker**, and Adalberto Simeone. **Comparison of Cross-Reality Transition Techniques Between 3D and 2D Display Spaces in Desktop–AR Systems**. In *IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 1–8, 2025.

[2] ★ **Matt Gottsacker**, Hiroshi Furuya, Zubin Datta Choudhary, Austin Erickson, Ryan Schubert, Gerd Bruder, Michael P. Browne, and Gregory F. Welch. **Investigating the relationships between user behaviors and tracking factors on task performance and trust in augmented reality**. In *Elsevier Computers & Graphics*, pages 1–14, 2024. doi:10.1016/j.cag.2024.104035.

Conference Publications

[3] ★ **Matt Gottsacker**, Yahya Hmaiti, Mykola Maslych, Hiroshi Furuya, Jasmine Joyce DeGuzman, Gerd Bruder, Gregory F. Welch, and Joseph J. LaViola Jr. **From One World to Another: Interfaces for Efficiently Transitioning Between Virtual Environments**. In *CHI Conference on Human Factors in Computing Systems (CHI)*, pages 1–17, 2026. doi:10.1145/3772318.3791912.

[4] Gerd Bruder, Ryan Schubert, Michael P Browne, Austin Erickson, Zubin Choudhary, **Matt Gottsacker**, Hiroshi Furuya, and Gregory F. Welch. **Exploring How Augmented Reality Display Features Affect Training System Performance**. In *International Conference on Human-Computer Interaction*, pages 75–95, 2025. doi:10.1007/978-3-031-93715-6_6.

[5] ★ **Matt Gottsacker**, Mengyu Chen, David Saffo, Feiyu Lu, Benjamin Lee, and Blair MacIntyre. **Examining the Effects of Immersive and Non-Immersive Presenter Modalities on Engagement and Social Interaction in Co-located Augmented Presentations**. In *CHI Conference on Human Factors in Computing Systems (CHI)*, pages 1–19, 2025. doi:10.1145/3706598.3713346.

[6] ★ **Matt Gottsacker**, Hiroshi Furuya, Laura Battistel, Carlos Pinto Jimenez, Nicholas LaMontagna, Gerd Bruder, and Gregory F. Welch. **Exploring Spatial Cognitive Residue and Methods to Clear Users' Minds When Transitioning Between Virtual Environments**. In *Proceedings of the IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 1000–1009, 2024. doi:10.1109/ISMAR62088.2024.00116.

[7] Hiroshi Furuya, Laura Battistel, Zubin Datta Choudhary, **Matt Gottsacker**, Gerd Bruder, and Gregory F. Welch. **Difficulties in Perceiving and Understanding Robot Reliability Changes in a Sequential Binary Task**. In *Proceedings of the ACM Symposium on Spatial User Interaction (SUI)*, pages 1–11, 2024. doi:10.1145/3677386.3682083.

[8] Juanita Benjamin, Austin Erickson, Matthew Gottsacker, Gerd Bruder, and Gregory F. Welch. **Evaluating Transitive Perceptual Effects Between Virtual Entities in Outdoor Augmented Reality**. In *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 619–629, 2024. doi:10.1109/VR58804.2024.00082.

[9] Hiroshi Furuya, Zubin Choudhary, Jasmine Joyce DeGuzman, **Matt Gottsacker**, Gerd Bruder, and Gregory F. Welch. **Using Simulated Real-world Terrain in VR to Study Outdoor AR Topographic Map Interfaces**. In *Proceedings of the International Conference on Artificial Reality and Telexistence Eurographics Symposium on Virtual Environments (ICAT-EGVE)*, pages 1–10, 2023. https://diglib.eg.org/server/api/core/bitstreams/7654d7af-6e9c-430c-b693-94a70aa0cfcc/content.

[10] **Matt Gottsacker**, Nahal Norouzi, Ryan Schubert, Frank Guido-Sanz, Gerd Bruder, and Gregory F. Welch. **Effects of Environmental Noise Levels on Patient Handoff Communication in a Mixed Reality Simulation**. In *Proceedings of the ACM Conference on Virtual Reality and Software Technology (VRST)*, pages 1–10, 2022. doi:10.1145/3562939.3565627.

[11] Nahal Norouzi, **Matt Gottsacker**, Gerd Bruder, Pamela J. Wisniewski, Jeremy Bailenson, and Gregory F. Welch. **Virtual Humans with Pets and Robots: Exploring the Influence of Social Priming on One's Perception of a Virtual Human**. In *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 311–320, 2022. doi:10.1109/VR51125.2022.00050.

[12] ★ **Matt Gottsacker**, Nahal Norouzi, Kangsoo Kim, Gerd Bruder, and Gregory F. Welch. **Diegetic Representations for Seamless Cross-Reality Interruptions**. In *Proceedings of the IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 310–319, 2021. doi:10.1109/ISMAR52148.2021.00047.

[13] Connor D. Flick, Courtney J. Harris, Nikolas T. Yonkers, Nahal Norouzi, Austin Erickson, Zubin Choudhary, **Matt Gottsacker**, Gerd Bruder, and Gregory F. Welch. **Trade-Offs in Augmented Reality User Interfaces for Controlling a Smart Environment**. In *Proceedings of the ACM Symposium on Spatial User Interaction (SUI)*, pages 1–11, 2021. doi:10.1145/3485279.3485288.

[14] Zubin Choudhary, **Matt Gottsacker**, Kangsoo Kim, Ryan Schubert, Jeanine Stefanucci, Gerd Bruder, and Gregory F. Welch. **Revisiting Distance Perception with Scaled Embodied Cues in Social Virtual Reality**. In *Proceedings of the IEEE International Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 788–797, 2021. doi:10.1109/VR50410.2021.00106.

Workshop Publications

[15] ★ **Matt Gottsacker**, Yahya Hmaiti, Mykola Maslych, Hiroshi Furuya, Gerd Bruder, Joseph J. LaViola Jr., and Gregory F. Welch. **XR-First Design for Productivity: A Conceptual Framework for Enabling Efficient Task Switching in XR**. In *Adjunct Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 524–529, 2025. doi:10.1109/ISMAR-Adjunct68609.2025.00106.

[16] Hiroshi Furuya, Jasmine Joyce DeGuzman, Zubin Choudhary, **Matt Gottsacker**, Gerd Bruder, and Gregory F. Welch. **How Does Presence Affect Trust in Simulated Autonomous Agents?** In *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 502–503, 2025. doi:10.1109/VRW66409.2025.00109.

[17] Laura Battistel, **Matt Gottsacker**, Gerd Bruder, Gregory F Welch, Riccardo Parin, and Massimiliano Zampini. **Chill or Warmth: Exploring Temperature's Impact on Interpersonal Boundaries in VR**. In *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 621–623, 2024. doi:10.1109/VRW62533.2024.00121.

[18] **Matt Gottsacker**, Mengyu Chen, David Saffo, Feiyu Lu, and Blair MacIntyre. **Asymmetric Immersive Presentation System for Financial Data Visualization**. In *IEEE Visualization and Visual Analytics (VIS) MERCADO Workshop*, pages 1–6, 2023. https://osf.io/cvqg9/?view_only=6a90ddcce54f45c0a86649821c609d84.

[19] **Matt Gottsacker**, Mengyu Chen, David Saffo, Feiyu Lu, and Blair MacIntyre. **Hybrid User Interface for Audience Feedback Guided Asymmetric Immersive Presentation of Financial Data**. In *Adjunct Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 199–204, 2023. doi:10.1109/ISMAR-Adjunct60411.2023.00046.

[20] Robbe Cools[+], **Matt Gottsacker**[+], Adalberto Simeone, Gerd Bruder, Gregory F. Welch, and Steven Feiner. **Towards a Desktop-AR Prototyping Framework: Prototyping Cross-Reality Between Desktops and Augmented Reality**. In *Adjunct Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 175–182, 2022. doi:10.1109/ISMAR-Adjunct57072.2022.00040.

[+] : Denotes equal contributions.

Other Publications (Posters, Demos, etc.)

[21] **Matt Gottsacker**, Yahya Hmaiti, Mykola Maslych, Hiroshi Furuya, Gerd Bruder, Gregory F. Welch, and Joseph J. LaViola Jr. **From Alt-Tab to World-Snap: Exploring Different Metaphors for Swift and Seamless VR World Switching**. In *Adjunct Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 943–944, 2025. doi:10.1109/ISMAR-Adjunct68609.2025.00260.

[22] ★ **Matt Gottsacker**, Nels Numan, Anthony Steed, Gerd Bruder, Gregory F. Welch, and Steven Feiner. **Decoupled Hands: An Approach for Aligning Perspectives in Collaborative Mixed Reality**. In *CHI Conference on Human Factors in Computing Systems (CHI) Extended Abstracts (Late Breaking Work)*, pages 1–8, 2025. doi:10.1145/3706599.3720219.

[23] ★ **Matt Gottsacker**, Gerd Bruder, and Gregory F. Welch. **rlty2rlty: Transitioning Between Realities with Generative AI**. In *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 1160–1161, 2024. doi:10.1109/VRW62533.2024.00374.

[24] **Matt Gottsacker**, Raiffa Syamil, Pamela J. Wisniewski, Gerd Bruder, Carolina Cruz-Neira, and Gregory F. Welch. **Exploring Cues and Signaling to Improve Cross-Reality Interruptions**. In *Adjunct Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 827–832, 2022. doi:10.1109/ISMAR-Adjunct57072.2022.00179.

[25] **Matt Gottsacker**. **[DC] Balancing Realities by Improving Cross-Reality Interactions**. In *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pages 944–945, 2022. doi:10.1109/VRW55335.2022.00323.

[26] **Matt Gottsacker**, Steven R. Gomez, Richard Skowyra, and Flavio Esposito. **Toward Effective Visualization of Network Identifier Bindings in a Software-Defined Network**. In *ACM Internet Measurement Conference (IMC)*, 2018.

Teaching

Teacher, Computer Programming, Marquette University High School — 2019 – 2020

- Introductory Computer Science class taught to grades 10-12 using the Java programming language. Designed the course curriculum to be interdisciplinary and creative through projects that allowed students to incorporate their interests from other classes. Brought in guest speakers professionally designing and building software at companies such as Epic and Rivian.

Teaching Assistant, CAP 5115 Virtual Reality Engineering, University of Central Florida — Spring 2025

- Created tutorial videos, provided technical and teaching support in office hours, graded assignments.

Teaching Assistant, CGS 3269 Computer Architecture Concepts, University of Central Florida — Spring 2025

- Provided teaching support in office hours, graded assignments.

Teaching Assistant, CAP 6110 Augmented Reality Engineering, University of Central Florida — Fall 2024

- Created tutorial videos, provided technical and teaching support in office hours, graded assignments.

Guest Lecturer, AI & U: Artificial Intelligence Made Easy (for retirees), taught by Dr. Cathy Bishop-Clark, Miami University — 2025

- “Generative AI and Ethical Issues”

Guest Lecturer, CAP 6119 Advanced Virtual Reality, taught by Dr. Carolina Cruz-Neira, University of Central Florida — 2024

- “Transitional Reality User Interfaces”

Guest Lecturer, CAP 5115 Virtual Reality Engineering, taught by Dr. Ryan McMahan, University of Central Florida — 2021

- “System Fidelity”
- “Haptics”

Guest Lecturer, UCF NASA SUITS Workshop, Marquette University High School — 2021

- As the Outreach Lead on the UCF NASA SUITS team, led an educational event at MUHS Hilltopper Robotics to demonstrate and discuss how XR technology is used for space exploration.

Students Mentored

- Ahinya Alwin, University of Central Florida, undergraduate student (Computer Science)
- Abraham Hernandez, University of Central Florida, undergraduate student (Computer Science)
- Carlos Jimenez, University of Central Florida, undergraduate student (Game Design)
- Nicholas LaMontagna, University of Central Florida, undergraduate student (Game Design)
- Inne Maerevoet, KU Leuven, Master’s student (Computer Science)

Professional Service

XR Future Faculty Forum (F3)

- With Cassidy R. Nelson and Niall L. Williams, co-created F3 to connect aspiring and current faculty in the XR field through tutorials, panels, and 1-1 mentorship sessions at the IEEE ISMAR and VR conferences. Co-organized six consecutive F3 meetings at IEEE ISMAR and IEEE VR from 2023–2026.

XR Student Researchers Community

- Created a Discord server to connect students researching XR technologies all around the world. Over 350 students worldwide discuss research, organize meet-ups, recruit participants for online studies, and socialize.

Open-Source Tools for the Research Community

- *Peer Reviewer Recommendation Tool.* Static client-side web app that recommends peer reviewers for academic submissions via semantic similarity over HCI conference proceedings (sentence-transformers MiniLM-L6-v2 corpus embeddings; in-browser query embedding via Transformers.js).

Journal and Conference Reviewer. Since 2022, regularly reviewed papers at the venues listed below. **Received eight special recognitions for writing outstanding reviews.**

- ACM SIGCHI
- IEEE ISMAR
- IEEE VR
- IEEE TVCG
- Frontiers in Virtual Reality

Professional Associations

- Member, Association for Computing Machinery (ACM)
- Member, Institute of Electrical and Electronics Engineers (IEEE)

Program Committee Positions

- *Program Committee Member*, ACM NordiCHI 2026
- *Program Committee Member*, EuroXR 2026
- *Program Committee Member*, ACM VRST 2026

Conference Chair Positions

- *XR Future Faculty Forum Chair*, IEEE ISMAR 2025
- *XR Future Faculty Forum Chair*, IEEE VR 2025

- *XR Future Faculty Forum Chair*, IEEE ISMAR 2024
- *XR Future Faculty Forum Chair*, IEEE VR 2024
- *Student Volunteer Chair*, IEEE VR 2024
- *XR Future Faculty Forum Chair*, IEEE ISMAR 2023

Other Conference Service Positions

- *Student Volunteer*, IEEE ISMAR 2024
- *Paper Session Chair*, "Virtual Humans, Collaboration, and Social Interaction 1," ACM Conference on Virtual Reality Software and Technology (VRST), 2022
- *Student Volunteer and Mentor*, IEEE Virtual Reality and 3D User Interfaces (VR), 2022
- *Student Volunteer and Mentor*, IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2021
- *Student Mentor*, IEEE Virtual Reality and 3D User Interfaces (VR), 2021
- *Social Media Chair*, International Conference on Artificial Reality and Telexistence (ICAT) & Eurographics Symposium on Virtual Environments (EGVE), 2020